

Global Airport Resilience Index: Towards a comprehensive understanding of air transportation resilience

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ABSTRACT

Estimating the vulnerability of airport outages on the air transportation system is an ongoing research challenge. While existing studies are predominantly focused on the analysis of the air-side airport network, with airports being nodes and links representing direct flights, in this study we propose the so-called Global Airport Resilience Index for worldwide airports which incorporates ground infrastructure as well as population distribution for the computation of an integrated resilience index that estimates the effects of airport disruptions on the entire system. Based on the Global Airport Resilience Index of airports, we can derive realistic assessment for airport resilience worldwide, where a more important airport has a higher index value. The inherent challenges in data management and computation are significant and require sophisticated solutions. Overall, we believe that our study provides novel insights into air transportation and airport resilience, by consideration of a more realistic resilience estimation measure.

1. Introduction

The air transportation system is of paramount significance to our societies, reflecting the technological advancements driven by societal needs and global interconnectedness (Arvis and Shepherd, 2011; Janić, 2019; Cheung et al., 2020; Zhang and Graham, 2020; Chen et al., 2024). Accordingly, the ability of air transportation to withstand and recover from disruptions is instrumental for ensuring an ordered connectivity of societies on our planet; a concept generally defined as resilience; see Zhou and Chen (2020), Poo et al. (2021), Chen et al. (2024), Shen et al. (2024), Xu et al. (2024), Nickdoost et al. (2024), Gao et al. (2024), Ma et al. (2024) for recent studies on (air) transportation resilience and Ganguly et al. (2018), Zhou et al. (2019b), Bi et al. (2023) for reviews transportation resilience and more general critical infrastructure resilience. There exist three broad categories of system disruptions in the literature on air transportation resilience analysis¹: Random failures, targeted attacks, and singletons. The term *random failures* refers to unforeseen (or even unpredictable) events that can occur within various components of the air transportation system (Dangut et al., 2021; Pan et al., 2021; Wang et al., 2023). These failures can range from minor technical glitches to more severe structural issues, including disruptive weather events, medical emergencies, bird strikes, technical equipment malfunctions, ash clouds, as well as congestion. Studies on this subject are rather rare in the literature, and often focus on the estimation of failure probabilities, in contrast to the downstream effects. The term *targeted attacks* refers to deliberate acts of sabotage aimed

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¹ Throughout this study, we follow the formal meaning of resilience as defined by Rose (2007): “The ability of a system to maintain functionality when disrupted, with particular focus on the speed at which the system returns to normal”.

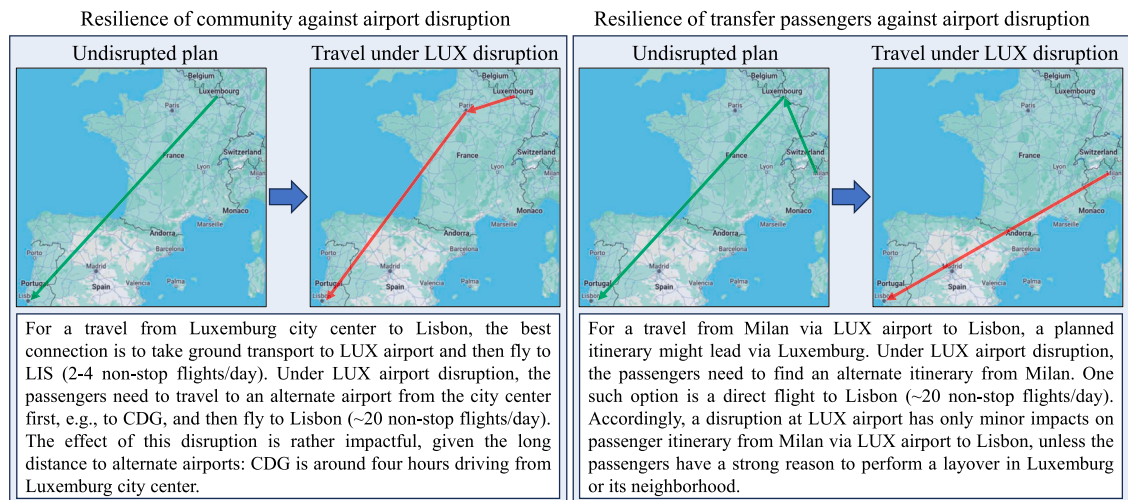


Fig. 1. Conceptual intuition for our Global Airport Resilience Index: community-perspective (left) and transfer perspective (right).

at system components (aircraft, airports, airlines or other infrastructure), through, e.g., cyber attacks, bomb threats, hijacking, or spoofing/jamming (Guimera et al., 2005; Verma et al., 2014; Sun et al., 2017a; Wandelt et al., 2021). These attacks are carried out with the sole intent to cause harm and disrupt air travel procedures, in order to achieve political/ideological goals or other criminal objectives. Naturally, such targeted attacks pose a significant threat to air transportation and require comprehensive countermeasures to mitigate the risks. Accordingly, this subject is predominantly investigated in the extant literature. Finally, the term *singleton* refers to an occurrence that happens only once within a given context; an event that is a singular instance and unlikely repeated or part of a series, including global financial crises or the COVID-19 pandemic. In the recent year, various studies have investigated the resilience of the air transportation system in face of the COVID-19 pandemic singleton (Bauranov et al., 2021; Janić, 2022; Li et al., 2023; Su et al., 2023).

In this study, we delve deeper into the mechanisms underlying air transportation resilience, aiming to elucidate the nature of resilience in the context of airport disruptions, better understanding the role of airport failures and its down-stream effects. In this context, our work is mostly related to the concepts of targeted attacks and singletons discussed earlier. Most existing work on this subject has performed comprehensive evaluations of airport disruptions using one of the following two types of approaches: (1) complex networks and (2) simulation/optimization-based endeavors. Studies based on complex networks often use nodal metrics to estimate node importance from multifaceted perspectives on a simplistic topological model, e.g., through degree or betweenness centrality. Simulation/optimization-based approaches – on the other hand – aim to model the disruption/recovery process as an optimization problem, usually under consideration of various real-world operational constraints. The intrinsics of these two methods, particularly their computational properties and data requirements, restrict their application to often oversimplified models (complex networks) versus small-scale instances (simulation/optimization). However, based on well-tuned simulation algorithms and data management, the void between the two former extremes can be filled, finely trading the extent of realism with the available computational resources.

The major contribution of our study is the proposition of the Global Airport Resilience Index (GARI),² a simulation-based approach, which measures the disruptive impact of airport closure on the entire air transportation system under consideration of two complementary perspectives; as visualized in Fig. 1. The first perspective of GARI is the effect of airport closure on the regional population; we refer to this effect as *community resilience*. Community resilience is measured in terms of an increase in passenger travel time under disruptions for starting/ending itineraries at an airport, with consideration of ground transportation and alternative flights. The second perspective of GARI is the effect of airport closure on the hub transfer activity of an airport; we refer to this effect as *transfer resilience*. The transfer resilience is measured in terms of an increase in passenger travel time to travel from their origin to destination under omission of the disrupted airport. The GARI index is evaluated based on global air transportation data, covering all single/multiple leg passenger itineraries between January 1st, 2023 and December 31st, 2023. Our experiments highlight the computational feasibility as well as the adequateness of GARI for estimating the resilience of the air transportation system against disruptions at individual airports. Through a comprehensive analysis of resilience-focused airport impact estimation techniques, we contribute to the discourse on strengthening the resilience of air transportation systems. Ultimately, with a comprehensive understanding of which resources to protect most, stakeholders can better mitigate risks, enhance operational efficiency, and foster a more sustainable and resilient air transportation ecosystem.

² The Global Airport Resilience Index (GARI) is inspired by the Global Airport Connectivity Index (GACI) proposed by Cheung et al. (2020).

It should be noted that there exists two different interpretations of the term *airport resilience index*. One interpretation comes with the connotation of how resilient a specific airport is towards the disruption of its components, e.g., blockage of a runway, some safety-related events in a terminal area or within the airport's catchment area (Malandri et al., 2017; Metzner, 2019; Postorino et al., 2020; Malandri et al., 2023; Zapola et al., 2024). The second interpretation refers to the importance of a specific airport within the air transportation network, e.g., how does the outage of an airport affect the airside or landside operations, e.g., how could passengers who normally pass through an airport could choose alternate itineraries for their travel or how can passengers inside the catchment area reach their destinations via other airports. This interpretation is the predominant one in the extant literature when it comes to resilience (see our literature review in Section 2). In our study, we are concerned with the second interpretation, measuring the impact of airport closure on the entire air transportation system under consideration of two complementary perspectives: community resilience and transfer resilience.

The remainder of this study is structured as follows. Section 2 reviews related existing studies on the resilience of air transportation, covering network science, and simulation/optimization-based methods. Section 3 describes the methodology for GARI and the underlying dataset. Section 4 evaluates GARI on data for the global air transportation system in year 2023. Section 5 concludes this study and provides suggestions for future work.

2. Literature review

There exist a significant number of studies that analyse the global air transportation system (Guimera et al., 2005; Verma et al., 2014; Sun et al., 2017a; Zhang et al., 2022) – or one of its subsystems – as a complex network. Given the vast number of studies on the subject, we referred to a few selected papers here and point interested readers to relevant surveys instead (Zanin and Lillo, 2013; Sun and Wandelt, 2021).

Studies related to the network science approach model the air transportation system as a network, with nodes representing airports and links representing, usually, the existence of direct flights between nodes. From this representation, a natural question is to identify the role of nodes in the networks and the importance of nodes from different perspectives. In this light, several existing studies apply complex network metrics to investigate the extent of criticality specific nodes have in the network representation. Fig. 2 highlights an example for the worldwide airport network with nodes marked according to their degree (number of airports reachable with direct flights). Here, the degree is a measure of local connectivity and in terms of resilience, this measure can be used to quantify the loss of connectivity incurred by airport closure. We summarize seminal studies below, in order of their appearance. Please note that some of these studies could also be attributed to the simulation/optimization-based category instead; see the discussion further below. In a pioneering study by Chi and Cai (2004), the U.S. airport network's response to targeted attacks was meticulously examined, focusing on various node topological attributes. Employing diverse node isolation strategies alongside a novel Bonacich power centrality-based approach, Lordan et al. (2014) revealed the disproportionate impact of isolating a small fraction of nodes on overall network functionality. Investigating the phenomenon of explosive percolation, Clusella et al. (2016) shed light on its implications within the U.S. airport network. Using spectral clustering, Cong et al. (2016) discerned critical airports within the Chinese airport network. Sun et al. (2017a) scrutinized the resilience of global airport networks under a spectrum of targeted attacks, employing diverse attack strategies and resilience metrics. Clark et al. (2018) conducted a comparative analysis of node properties within the U.S. air transportation network, elucidating their central roles in network robustness and recovery. Zhou et al. (2019a) investigated were one of the first researchers to investigate the efficiency and robustness of weighted air transport networks. Clemente and Cornaro (2019) introduced the effective resistance centrality measure, based on the relative drop of the Kirchhoff index, and evaluated its applicability within the U.S. air transportation network. Wong et al. (2020) adopted a multifaceted approach to analyse the resilience of individual U.S. airline networks, advocating for a holistic understanding of network robustness. Charting the evolution of the Chinese air transportation city network's robustness, Chen et al. (2020) offered insights spanning from 1975 to 2017. Sun et al. (2020) identified pivotal nodes within air transportation city networks. Zhang et al. (2021) scrutinized the robustness of air transportation networks in the belt-and-road region, employing advanced complex network techniques. Overall, studies in this category either use node importance measure directly or aim to develop alternate techniques dismantling the network based on mostly topological indicators.

In recent years, more simulation/optimization-based approaches have been proposed. Wandelt et al. (2015) introduced a node importance assessment methodology for complex air transportation networks, leveraging Monte Carlo tree search techniques and contrasting findings against conventional network metrics across four distinct aviation networks. Requião da Cunha et al. (2015) devised a module-based attack framework rooted in topological communities, evaluating its efficacy within the U.S. airport network. Using a dataset on passenger itineraries flown during February 2013, Voltes-Dorta et al. (2017) simulated the full-day individual closures of the 25 busiest European airports and disrupted passengers then relocated to minimum-delay itineraries. The resulting aggregated delays are used to derive an airport importance ranking. Wang et al. (2019a) were the first to estimate the impact of partial airport outage by developing a framework consisting of a queuing model and delay propagation algorithm. Through experiments on the Chinese air transportation network, critical airports were identified which influence the performance of whole system; and results compared to topological network metrics. Wandelt et al. (2021) proposed a community-based complex network attack strategy, assessing its efficacy, among others, on air transportation systems. Malandri et al. (2023) took a different perspective by experimenting with variations of selected key performance indicators for baseline and disrupted conditions. The methodology is applied to European airports. Esteve et al. (2023) investigated the resilience of the European air route network between 2015 and 2018, with several deactivation strategies, covering link deactivation and region (grid cell) deactivation. Wandelt et al. (2023b) proposed a mixed-integer programme formulation for an airline recovery baseline under node disruptions and compared the results against topological indicators. It is shown that existing simplifications based on complex networks often underestimate the effect of node failures and that there exist significant ranking mismatches, particularly for top-ranked nodes.

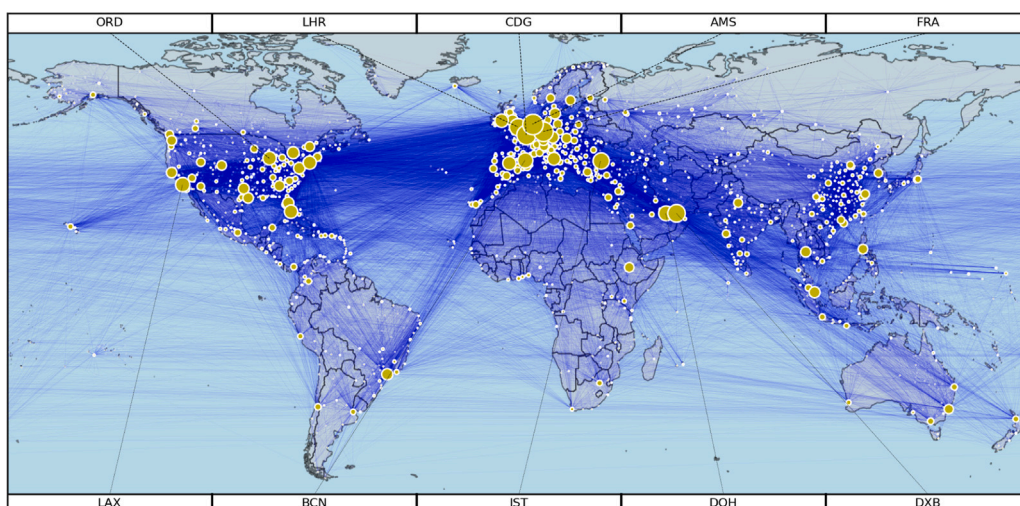


Fig. 2. Global network with nodes ranked according to their degree (number of airports connected by direct flights) for the year 2023. Direct flights are visualized with blue links and airports with yellow circles; the size of a circle correlates with the degree of an airport. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
List of major airport codes for Western Europe.

IATA code	Airport name	Municipality
LHR	London Heathrow Airport	London
CDG	Charles de Gaulle International Airport	Paris
AMS	Amsterdam Airport Schiphol	Amsterdam
FRA	Frankfurt am Main Airport	Frankfurt am Main
MAD	Adolfo Suárez Madrid-Barajas Airport	Madrid
BCN	Josep Tarradellas Barcelona-El Prat Airport	Barcelona
MUC	Munich Airport	Munich
LGW	London Gatwick Airport	London
FCO	Rome-Fiumicino Leonardo da Vinci International Airport	Rome
DUB	Dublin Airport	Dublin

3. Measuring the importance of airports

Section 3.1 provides a conceptual overview on the problem of airport resilience estimation. Section 3.2 introduces the formal resilience estimation framework based on air-ground passenger itineraries. Section 3.3 presents how the framework is used to compute GARI for the local population. Section 3.4 discusses how the framework is used to compute GARI for the transfer passengers.

3.1. Overview

In order to explain the methodology of GARI, we temporally use a subset of the global air transportation system in this section. The major rationale for a discussion on a smaller example is the significantly reduced number of nodes in the transportation network and, accordingly, a clearer presentation of the intuitions behind definitions and algorithms. We have chosen an area corresponding to Western Europe; see Fig. 3. Airports are visualized with black circular markers and direct flights represented by links in magenta colour. The width of a link corresponds to the number of passengers transported in the year 2023. We can observe a densely connected network of direct flights. The top 10 airports according to the number of passengers are annotated with their IATA code. Table 1 summarizes metadata of these airports with their IATA code, airport name, and associated municipality.

As highlighted in Section 2, many existing studies investigate the air-side connectivity of airports from a topological perspective in order to estimate the importance of an airport, i.e., quantifying the effect of the disruption of the airport on the network's connectivity. Fig. 4 visualizes airports' betweenness centrality (the number of shortest paths an airport appears on). Few nodes in the network have a relatively high betweenness centrality, e.g., PMI (Palma de Mallorca Airport), CDG, and BCN, indicating that these airports would be passed frequently by a random traveller. While such a topological perspective has merits and offers valuable insights into the structural characteristics of airport networks from a network science perspective; topological metrics exhibit significant limitations when employed in isolation for assessing airport importance under disruptions. For instance, such measures often overlook crucial factors inherent to dynamics and operational characteristics, such as the actual volume of passenger (or cargo traffic) flowing through airports, which might not necessarily align with their topological centrality derived on all-pairs-shortest paths, as in case of betweenness centrality. Moreover, the exclusive focus on direct connections neglects the importance

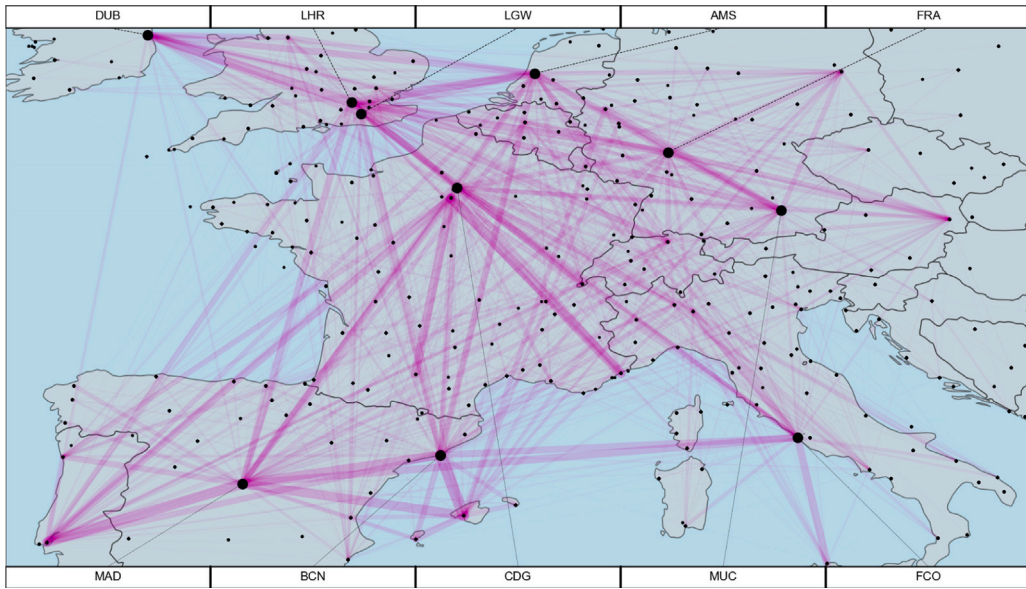


Fig. 3. Overview on airport connectivity in Western Europe.
Source: The data is from Sabre Market Intelligence.

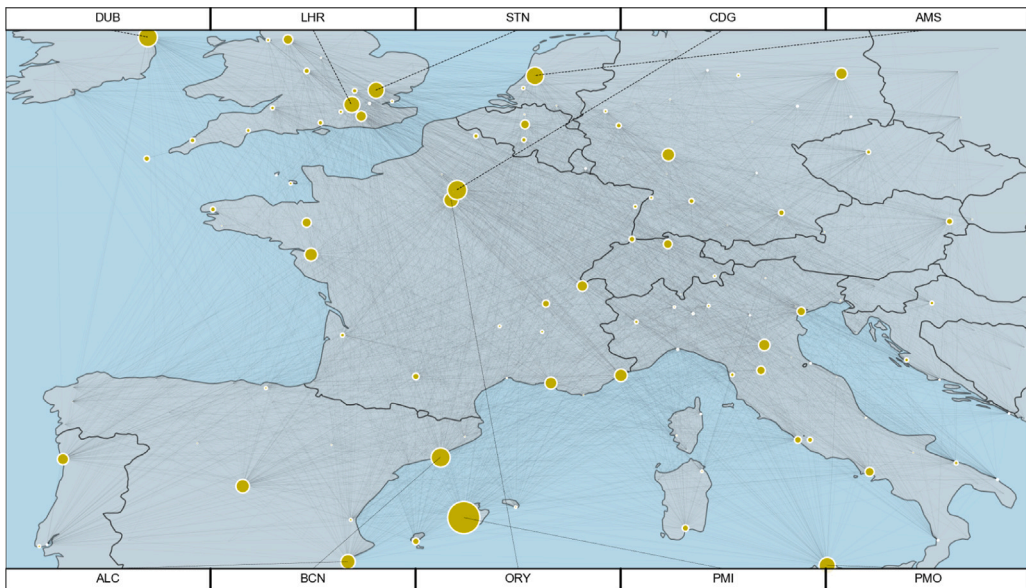


Fig. 4. Betweenness centrality of airports, based on airport connections inside Western Europe.

of transit hubs and indirect connections in facilitating air travel. Being complex systems, the analysis of airport network resilience demands a more comprehensive approach that integrates not only topological metrics but also considerations of geographic location, accessibility, and passenger itineraries to provide a nuanced understanding of airport importance. Relying solely on topological measures risks oversimplifying this complexity and might result in misleading policy decisions and inadequate air transportation infrastructure protection.

The example of airport Palma de Mallorca Airport (PMI), the airport with the highest betweenness centrality in Western European, is rather instructive. While PMI serves as a touristic hub in the Balearic Islands, its importance in the broader context of air transportation is relatively limited: Unlike major international hubs such as London Heathrow (LHR) or Amsterdam Schiphol (AMS),

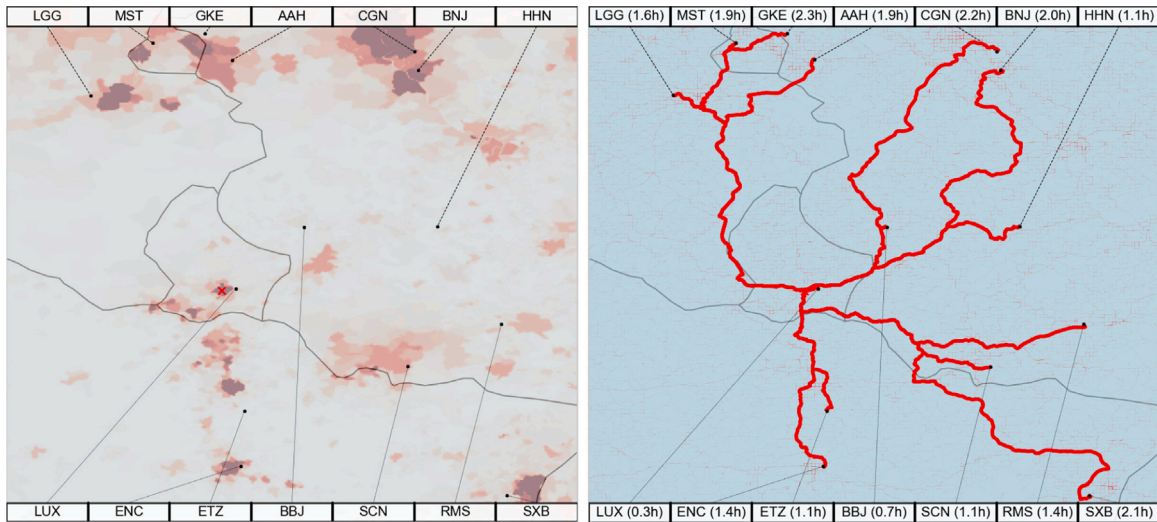


Fig. 5. Grid-based population distribution and airports around Luxembourg city centre (left) and driving times/routes from Luxembourg city centre to nearby airports (right).

PMI does not play a pivotal role in connecting multiple continents or serving as a crucial transit point for long-haul flights. Second, PMI's geographic location and route network primarily cater to leisure travellers visiting Mallorca, Ibiza, and Menorca. While tourism is undoubtedly vital for the local economy, it does not necessarily translate into being essential for air transportation on a wider scale. Furthermore, the presence of alternative airports in the region, such as Barcelona-El Prat Airport (BCN) and Madrid-Barajas Adolfo Suárez Airport (MAD), could provide additional options for travellers to access the Balearic Islands, potentially reducing the dependence on PMI for air transportation. In summary, the betweenness of PMI potentially overestimates the importance of PMI in the Western European air transportation system, simply given the number of direct flights connecting to PMI, particularly in summer.

Another interesting example is Luxembourg-Findel International Airport (LUX) - an airport with one of the smallest betweenness values in Fig. 4; note that LUX is not annotated with a label, since it does not belong to the top-10 airports. While having a negligible betweenness value, LUX serves as a hub for both passenger and cargo traffic in Europe, with a particular focus on the latter. As the only international airport in Luxembourg, it plays a pivotal role in connecting the country to destinations across Europe and beyond. Fig. 5 shows the population distribution around Luxembourg airport (left) and the driving times/routes from Luxembourg city centre to nearby airports (right). The connectivity through LUX is essential for facilitating business travel and trade, contributing to Luxembourg's status as a global financial and logistics centre. Moreover, LUX's strategic location in the heart of Europe makes it an ideal transit point for flights between Asia, the Americas, and other parts of the world; a benefit which strongly materializes for cargo transportation, underscoring its importance in global supply chains. In summary, the betweenness of LUX potentially underestimates the role for the local population. In this context, it should be noted that LUX is well-connected to several hubs nearby, who are also connected among themselves, explaining the relatively low betweenness centrality values (see Fig. 6).

Given these two examples, it becomes clear that a realistic estimation of airport resilience needs to go beyond simplistic topological properties. For the definition of GARI, we consider two complementary resilience perspectives. The first perspective concerns the so-called *community* aspect, i.e., how does an airport enhance connectivity by providing access to domestic and international destinations. Such a community resilience is important, since a connectivity highly correlates with investment, competitiveness of local industries, cultural exchange and tourism. We refer to this community-based measure as *GARICommunity*. The second perspective concerns the *transfer* aspect, i.e., how does an airport provide distribution of transfer passengers, who change aircraft at the airport to reach their destination. Such a transfer resilience is important, since hubs are often strategically located and well-connected, serving as central points where passengers from various origins can connect to flights heading to diverse destinations. The distribution of transfer passengers generates significant economic benefits for airports, airlines, and the local economy. We refer to this measure as *GARITransfer*. Both measures, *GARICommunity* and *GARITransfer*, are highly complementary in nature and will be introduced below.

3.2. Itineraries in the air-ground network

This section defines the mathematical preliminaries for our two GARI indicators. Specifically, we define an integrated air-ground network which is used for routing air passenger itineraries under disruptions. Let I be a set of itineraries, such that each itinerary $i \in I$ is a tuple (o_i, d_i, p_i, r_i) where:

- o_i represents the origin of the itinerary i ,

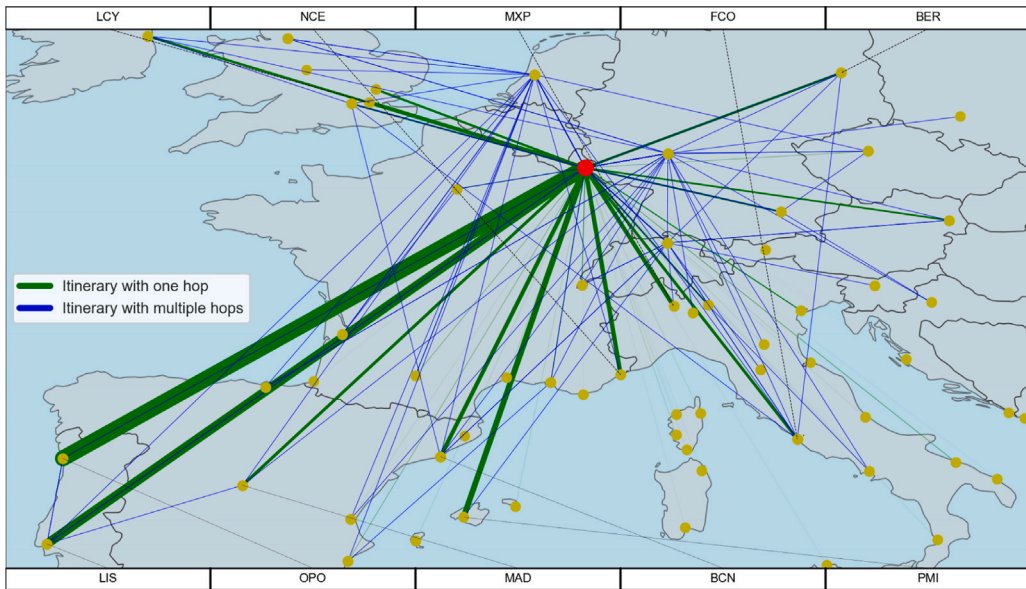


Fig. 6. OD itineraries originating in Luxembourg, distinguishing single-hop itineraries (green) and multiple-hop itineraries (blue). The line width correlates with the number of passengers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- d_i represents the destination of the itinerary i ,
- p_i represents the number of passengers taking the itinerary i ,
- r_i represents the route of itinerary i , consisting of a sequence of n nodes $r_i = [x_i^1, \dots, x_i^n]$. The first element x_i^1 , corresponds to o_i and the last element x_i^n corresponds to d_i .

We distinguish two types of itineraries in this study. The first type of itinerary is called *air-side itinerary*, denoted by I^{air} , where each x_i^j consists of an airport. The set of air-side itineraries can be obtained via existing aviation data aggregators. We use Sabre Market Intelligence,³ a commercial data source. Notably, Sabre Market Intelligence provides data at a worldwide scale, allowing us to derive a global resilience index. From Sabre Market Intelligence, we obtained a collection of passenger OD (origin–destination itineraries), including intermediate stops. While only about 20% of these itineraries consist of two airports (representing a direct flight from origin to destination), these direct itineraries cover about 85% of all passenger trips worldwide. In this study, we take all air-side itineraries I^{air} as a foundation for the passenger flow in the undisrupted global air transportation system. An air-side itinerary i becomes disrupted if one of the airports in r_i is disrupted. In case of such a disruption, we aim to find a feasible alternate itinerary and then compute the estimated impact in order to derive an airport resilience indicator. Defining such an airport resilience indicator without consideration of ground transportation would lead to limited insights, as passengers clearly can use alternate airports (Wang et al., 2019b) for access/egress, particularly in multiple airport regions; and the efforts for reaching alternate airports should be taken into account.

Accordingly, we need to extend the definition of airside itineraries to *air–ground itineraries* in order to capture passenger choice in real-world transportation. Intuitively, the major difference to air itineraries is that r_i can now also contain nodes on the ground, mainly covering the catchment area of airports. Following related studies on multiple airport regions (Sun et al., 2017b), we take a threshold of 1.5 h driving time as a cut-off to compute the relevant catchment area around an airport. Specifically, we use the Gridded Population of the World⁴ at 2.5 min as resolution for input. Each grid cell in this dataset is annotated with the latitude/longitude and population density of the cell.⁵ Fig. 7 visualizes the obtained grid cell population data for the example area in this section.

In addition, we use Openstreetmap to extract the worldwide ground transportation infrastructure; see Wandelt et al. (2023a) for details of the network extraction procedure; see Fig. 8 for an extract on Western Europe. We map the population grid cells to their closest ground infrastructure and compute the driving time to airports. The driving time is obtained by computing free-flow travel times respecting max speed limits from ways in Openstreetmap. Following this preparation technique, we obtain for each airport a population-weighted catchment area grid with realistic driving times and driving routes from the airport. We denote the nodes belonging to the population-weighted catchment area of an airport ap with $catch(ap)$. Fig. 9 visualizes the obtained catchment areas of three airport reporting colour-coded driving times (top) and population weights (bottom).

³ <https://www.sabre.com/products/suites/data-and-analytics/sabre-intelligence-exchange-ix/>

⁴ <https://sedac.ciesin.columbia.edu/data/collection/gpw-v4>

⁵ Technically speaking, the dataset is stored in GeoTIFF format, which allows to extract population densities from specific latitude/longitude pairs.

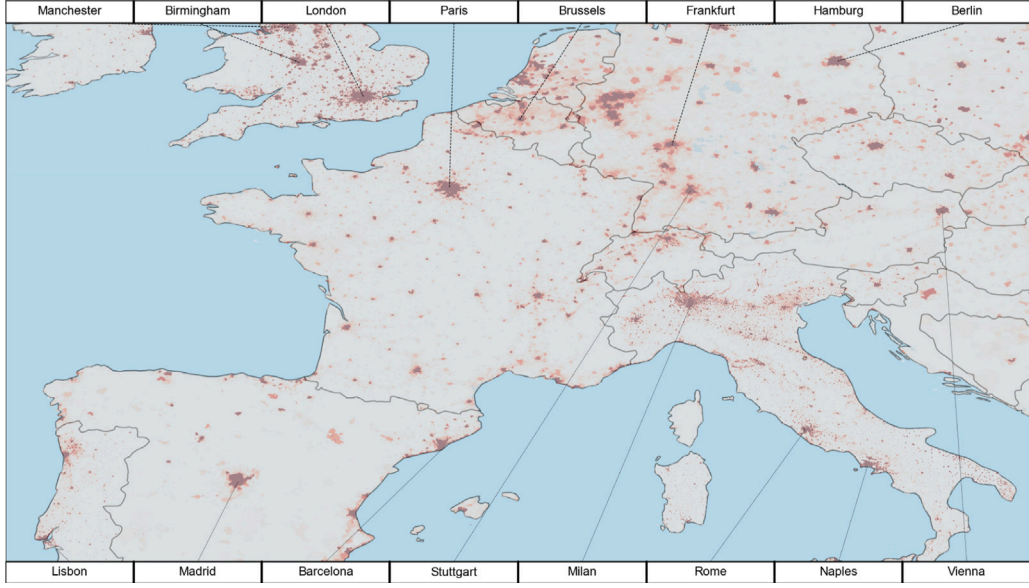


Fig. 7. Population distribution and major municipalities in Western Europe.

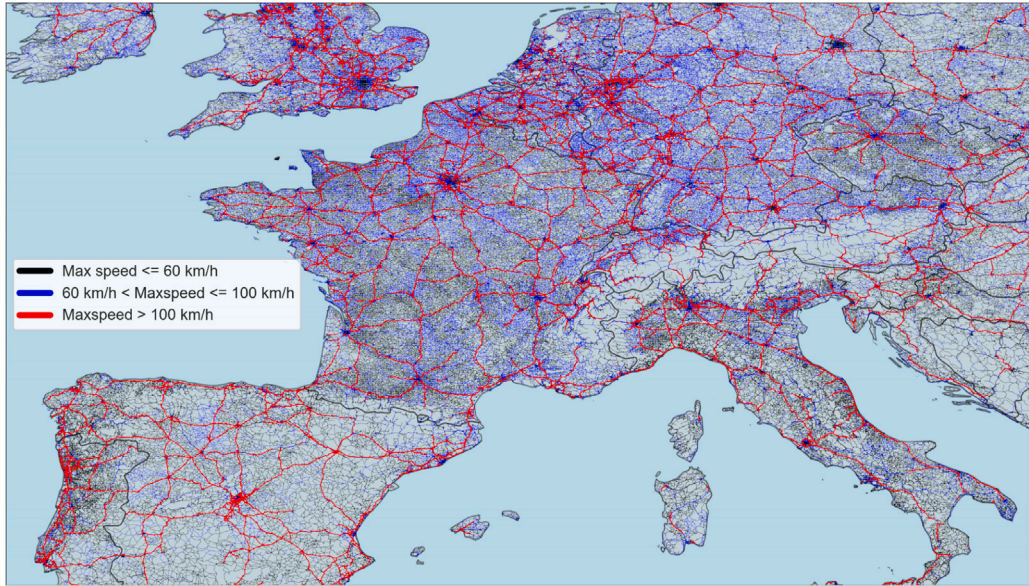


Fig. 8. Ground network extracted from Openstreetmap, with links colour-coded according to maximum speeds.

In the next step, we extend the air-side itineraries I^{air} to air-ground itineraries $I^{air,ground}$. There are two major differences between I^{air} and $I^{air,ground}$. First, the r_i sequence does not only contain airports as elements but also ground nodes. Second, the passengers in $I^{air,ground}$ are weighted according to the relative population density of the grid cell and distance from the airport, compared to all grid cells in the population-weighted catchment area of the airport, since we cannot determine the actual start of a passenger itinerary on the ground transportation network. We make a probabilistic estimation based on the airport catchment area around the airport instead. Specifically, we convert an air-side trajectory $i = (o_i, d_i, p_i, r_i) \in I^{air}$ into a set of air-ground itineraries i^{ag} as follows. Let $CP = catch(o_i) \times catch(d_i)$, i.e., the cross product of the nodes in the two catchment areas of the origin airport and destination airport, respectively. Then for each $(j, k) \in CP$, we create one air-ground itinerary $jk = (o_{jk}, d_{jk}, p_{jk}, r_{jk})$, such that:

- $o_{jk} = j$, the origin node on the ground,

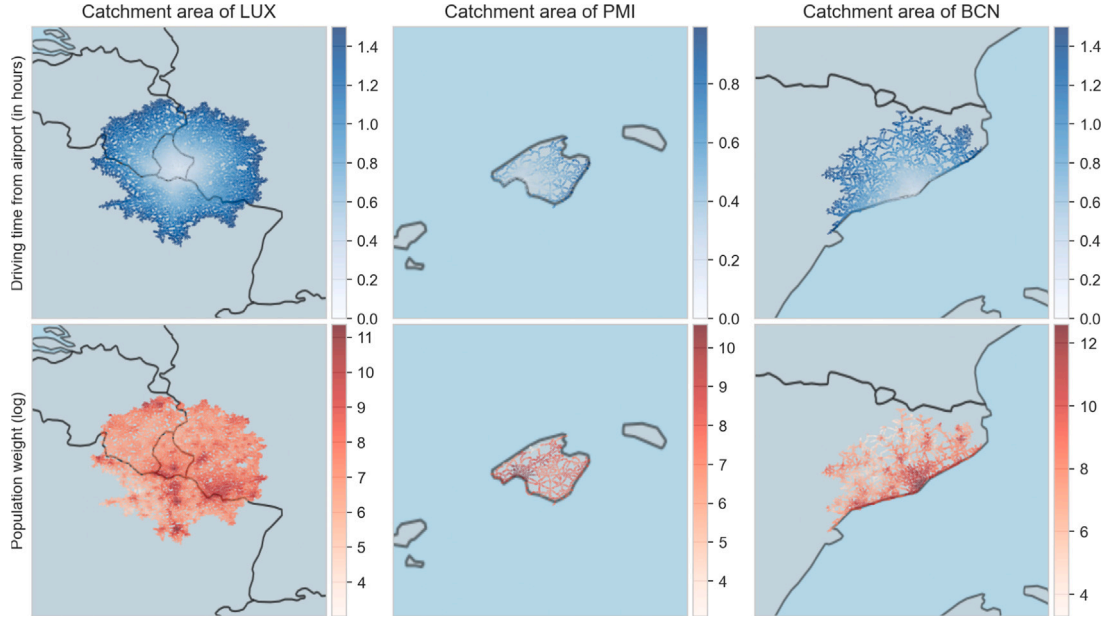


Fig. 9. Three example catchment areas based on driving time-filtering (top) and with population weight (bottom).

- $d_{jk} = k$, the destination node on the ground,
- $p_{jk} = p_i * \frac{f(j, o_i) * f(k, d_i)}{\sum_{l \in \text{catch}(o_i)} f(l, o_i) * \sum_{m \in \text{catch}(d_i)} f(m, d_i)}$ represents the estimated number of passengers taking the itinerary, where $f(j, ap)$ represents the relative demand for cell j regarding airport ap , computed as $\frac{\text{Norm}(\text{popd}(j))}{\text{Norm}(\text{dist}(j, ap))}$ with $\text{popd}(j)$ being the population density at grid cell j , $\text{dist}(j, ap)$ being the driving distance between j and ap according to the extracted ground infrastructure from Openstreetmap (in km), and Norm is the Min-Max normalization⁶ for grid cells at origin and destination, respectively,
- r_{jk} is the concatenation of the shortest route from j to o_i , the air-side route r_i , and the shortest route from d_i to k , where (o_i, d_i, p_i, r_i) is the input air-side itinerary.

It should be noted that the sum over all p_{jk} is equal to p_i , as ensured through the relative weighting population density and distance from an airport within a catchment area. Accordingly, our transformation from air-side itineraries I^{air} to air-ground itineraries $I^{\text{air,ground}}$ leads to a probabilistic representation of all trajectories between the airports' grid cells weighted by their relative contribution of population, i.e., two grid cells with larger population density will contribute a larger fractional flow p_{jk} in the corresponding air-ground directory. Fig. 10 visualizes this transformation. In this example, the original air-side example is the route LUX, AMS, MAD, i.e., passengers flying from Luxemburg to Madrid via Amsterdam. The catchment areas for LUX (origin) and MAD (destination) are highlighted in blue and green colour, respectively. The ground route from j to LUX and from MAD to k is highlighted in yellow colour, respectively.

To sum up, each air-side itinerary encodes flow between a pair of airports. These the itineraries represent known origin-destination data between worldwide airports as represented in Sabre Market Intelligence. We then dissect the observed flow along air-side trajectories into probable air-ground itineraries, based on population densities and distances to the airports, following the intuitions of a simple gravity model in each airport's catchment area. The major constraint of our transformation from a single air-side itinerary (o_i, d_i, p_i, r_i) into a set of air-ground itineraries is that the sum of passengers along the set of obtained air-ground itineraries is p_i , i.e., the overall flow observed in the real data from Sabre Market Intelligence is preserved. Due to this normalization, there is no overestimation of passengers, since the sum of the number of passengers on all air-ground itineraries for an given airside itinerary is equal to p_i .

Having extended the notion of air-side itineraries (between airports) to air-ground itineraries (between populated grid cells), we next describe a way to compute a ground itinerary's average estimated travel time. This computation will be essential for approximating the effect of airport closure on the local population and transfer passengers. Our goal is to be able to reduce the identification of alternate trajectories to a graph-based shortest-path-like problem which can be solved efficiently even for a dataset on a global scale. Accordingly, the key for developing travel time measure is to derive appropriate edge and node weights. Edge weights represent the time spent on an edge for travelling. For the ground network, we can simply use the maximum speed from Openstreetmap to derive travel times along edges; in absence of a speed limit, we use 40 km/h driving speed as an estimation during routing. For the air-side edges (direct flights) we have used the average scheduled flight time between each airport pair, obtained

⁶ See the Appendix for a discussion of different normalization choices.

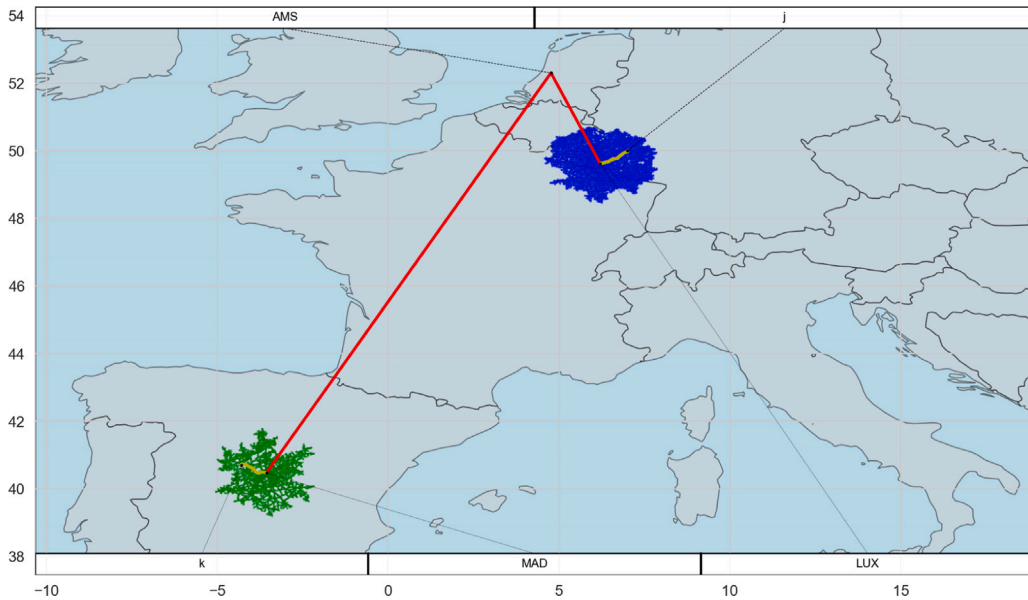


Fig. 10. Example visualization of an air-ground itinerary between j and k for the air-side itinerary from LUX, via AMS, to MAD.

from Sabre Market Intelligence, as a travel time. We add an overhead to the flight time considering two aspects. First, we have set the time for boarding and security check-in purposes to one hour. Second, based on the daily number of departures between the airport pair, we have derived a destination-specific average waiting time; since we do not consider the time of the day. Taking into account the time of the day would increase the degree of realism, but it turned out to be too complex computationally, not allowing us to solve hundreds of thousands of large-scale routing problems for the planet. Accordingly, integrating the time of day is a very interesting, yet challenging direction for future work.

3.3. *GARICommunity: Measuring resilience for the local population*

The idea underlying the computation of *GARICommunity* is to estimate the increase in travel times for the local population (i.e., community) around an airport once this airport becomes disrupted. Our disruption scenario assumes that an airport is entirely closed,⁷ which means that the local population has to find alternate ways to reach their destination. For such alternate itineraries, we consider interactions between air and ground transportation. Specifically, given that the airport is closed, the local population has to use an alternate airport to start their travel, unless using ground transportation alone leads to the best alternate itinerary through the air-ground network. Accordingly, in this study, we assume that the passengers predominantly make choices based on travel time; while there exist other criteria involved in the decision process, such as money or airline (mile programmes). Nevertheless, in case of emergency conditions, it is plausible to assume that passengers who want to travel mainly are concerned with the ability to reach the destinations in time. Below, we describe the detailed procedure for computing *GARICommunity*.

For each airport ap , we perform the following steps. First, determine air-ground itineraries which pass ap as the origin airport and denote these itineraries as U . This gives us a baseline for estimated population-weighted travel times from the airport catchment area of ap to all destinations in an undisrupted scenario. Next, we temporarily remove all air edges involving ap from the current air-ground network, essentially forbidding any flights via the airport ap . Then, for each air-ground trajectory in $jk \in U$, we (re)compute the shortest path from o_{jk} to d_{jk} in the disrupted network, i.e., avoiding the usage of ap , obtaining the recovered itineraries R . Finally, we compute the travel times for all air-ground itineraries without using ap , we denote the obtained travel time difference (recovered travel time minus undisrupted travel time) of recovered itinerary jk with $tt_d(jk)$. We consider itineraries with travel time increases only to avoid the following problem: Some of the initial air-side itineraries are not based on travel time considerations. Therefore, the travel time in a disrupted network could even be shorter than in an undisrupted network. Similarly, for nearby airports, it might be that ground transportation is faster than a disrupted flight leg. Since we are interested in the negative impacts

⁷ This assumption is predominant in the extant literature, particularly for works in the network science domain. Developing a stochastic resilience measure which takes into consideration partial closures is an important future research direction. Nevertheless, at the planet scale, this presumably adds an order of magnitude time complexity for performing all experiments for airports worldwide. Accordingly, we conjecture that more sophisticated routing algorithms and some simplifications will be required.

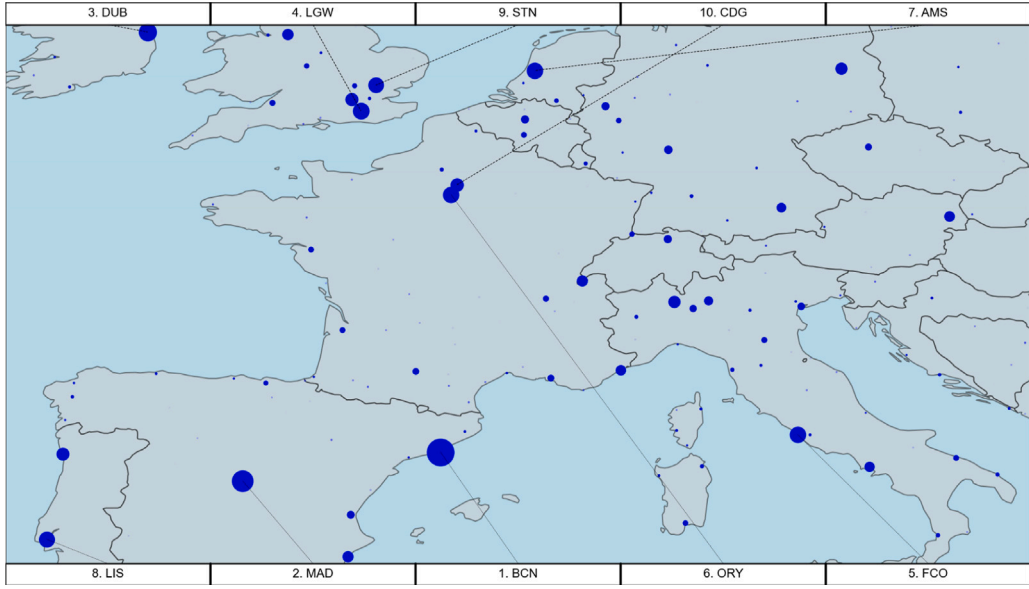


Fig. 11. GARIcommunity values for Western European airports, with Top 10 airports highlighted by IATA code.

of disruptions, we consider only these itineraries where the disrupted travel time is longer than the undisrupted travel time. Finally, we let $GARIcommunity(ap) = \sum_{\{jk \in U | ttd(jk) > 0\}} p_{jk} * ttd(jk)$.

Fig. 11 visualizes the values of *GARIcommunity* for the example region. We can see that the three top-ranked airports, BCN, MAD, and DUB, are all located outside the catchment areas of other airports with significant traffic. Accordingly, once such an airport gets disrupted, the local population does not have much choice for alternate routes. In some other cases, for instance ORY, there might be an alternate airport nearby, i.e., CDG, but with significantly different destination profiles. Finally, it should be noted that PMI did not appear among the top 10 airports mainly because the local population around the airport who travels from there is significantly smaller compared to other European airports.

3.4. GARItransfer: Measuring resilience towards passenger transfer functions

The idea underlying the computation of *GARItransfer* is to estimate the increase in travel times for the transfer passengers once an airport becomes disrupted. Our disruption scenario assumes that an airport is entirely closed. Notably, we do not consider the case that the origin is equivalent to the disrupted hub; as this case has been discussed in *GARIcommunity*. For alternate itineraries, we again consider interactions between air and ground transportation. Given that the airport is closed, we compute the route from the origin of the air-ground itineraries passing through the disrupted airport to the destination of the air-ground itinerary. Below, we describe the detailed procedure for computing *GARItransfer*.

For each airport ap , we perform the following steps. First, determine air-ground itineraries which pass ap as a connecting airport, i.e., not as the first airport or the last airport in the itinerary; we denote these itineraries as U . This gives us a baseline for estimated population-weighted travel times for transfer passengers through ap in an undisrupted scenario. Next, we temporarily remove all air edges involving ap from the current air-ground network, essentially disallowing any flights via the airport ap . Then, for each air-ground trajectory in $jk \in U$, we (re)compute the shortest path from o_{jk} to d_{jk} in the disrupted network, i.e., avoiding the usage of ap , obtaining the recovered itineraries R . Finally, we compute the travel times for all air-ground itineraries without using ap , we denote the obtained travel time difference (recovered travel time minus undisrupted travel time) of recovered itinerary jk with $ttd(jk)$. Similarly to *GARIcommunity*, we consider itineraries with travel time increases only. Finally, we let $GARItransfer(ap) = \sum_{\{jk \in U | ttd(jk) > 0\}} p_{jk} * ttd(jk)$.

Fig. 12 visualizes the values of *GARItransfer* for the example region. It can be seen that a few airports stand out, with significantly larger *GARItransfer* values compared to the other airports. The three top-ranked airports, MAD, FRA, and CDG, are all major European transportation hubs, indicating the appropriateness of our definition for *GARItransfer*. Most other airports' *GARItransfer* values are relatively small, indicating a high resilience of the air transportation system's transfer passengers against disruptions at these airports. Nevertheless, these results should be understood in context of the spatial subregion, as we did not include flights starting or landing outside the region for this example.

4. Experimental evaluation at planet scale

In this section, we evaluate the two airport resilience indicators, *GARcommunity* and *GARItransfer*, on a planet-scale dataset in the year 2023. The following data sources have been used to construct a global air-ground trajectory dataset:

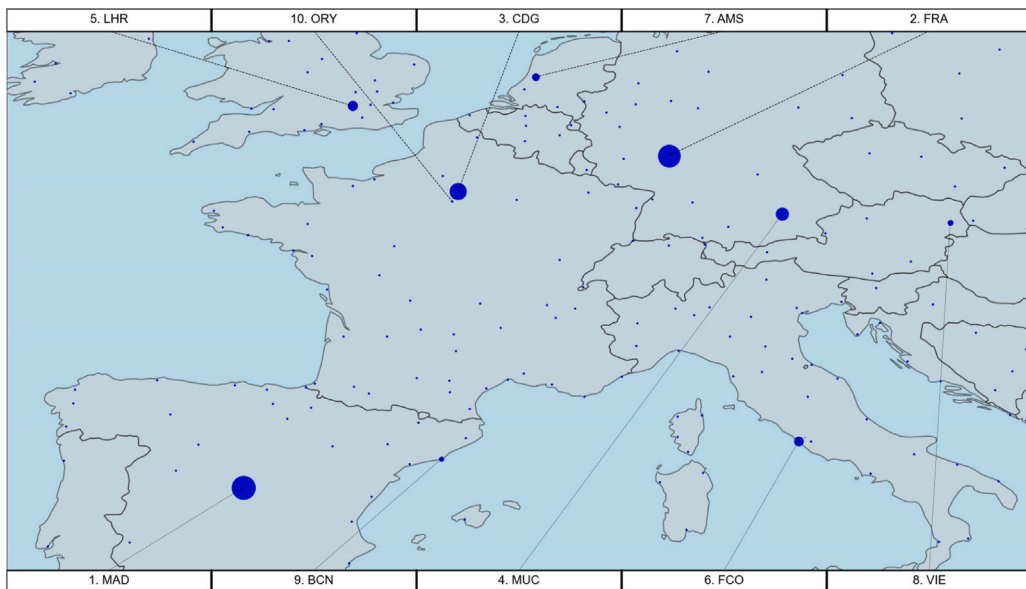


Fig. 12. GARItransfer values for Western European airports, with Top 10 airports highlighted by IATA code.

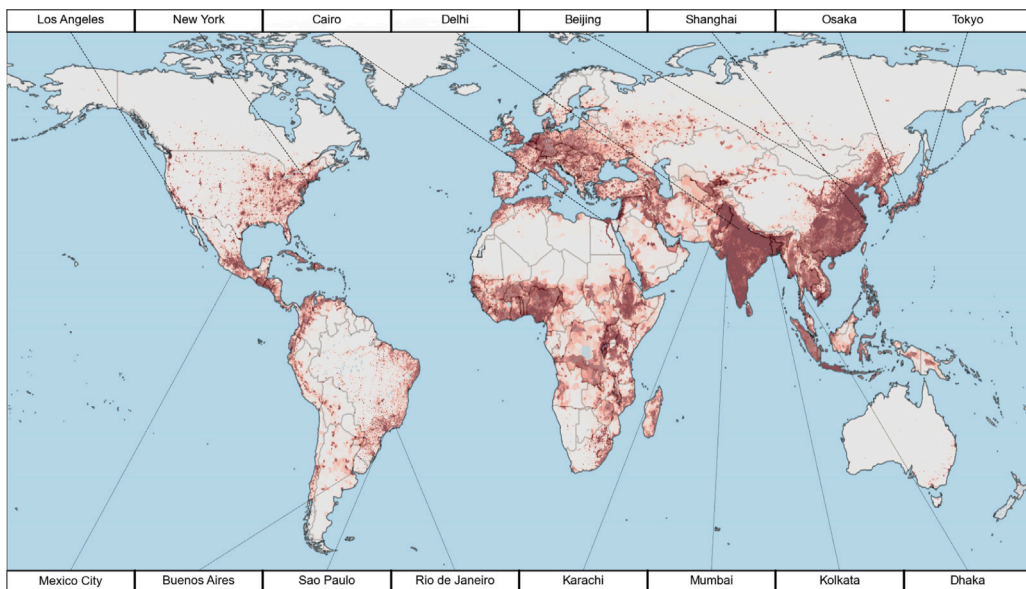


Fig. 13. Gridded Population of the World and top 16 cities according to population size.

- Sabre Market Intelligence⁸: We have obtained all direct and indirect passenger itineraries worldwide from January 1st, 2023 to December 31st, 2023. Each itinerary is annotated with the number of passengers that took the itinerary.
- Openstreetmap⁹: We have extracted the ground infrastructure network based on the ways annotated with *highway* attributes. The speed maximum of each way was extracted; we used 40 km/h in absence of a value.
- Gridded Population of the World¹⁰: Population distribution data at grid level was extracted from GeoTIFF files. Fig. 13 shows a visualization of the grid.

⁸ <https://www.sabre.com/products/suites/pricing-and-revenue-optimization/market-intelligence/>

⁹ <https://www.openstreetmap.org>

¹⁰ <https://sedac.ciesin.columbia.edu/data/collection/gpw-v4>

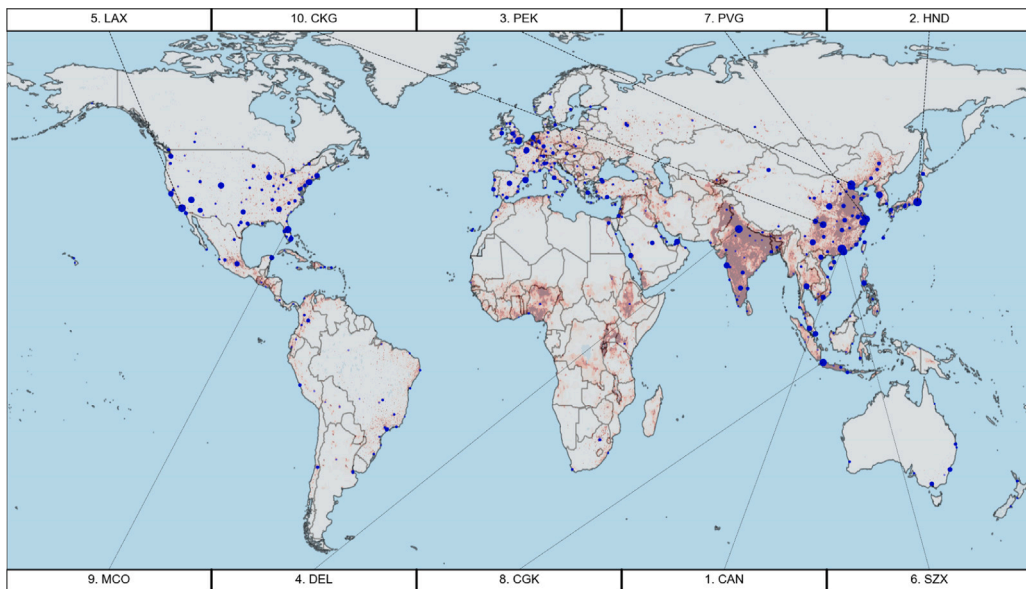


Fig. 14. *GARIcommunity* for the planet.

We report the *GARIcommunity* values from our planet-wide resilience analysis in Fig. 14. The most important airport for the local population is HND (Haneda Airport, also called Tokyo International Airport). While HND is one of the two international airports serving the Greater Area of Tokyo, the other one being NRT (Narita International Airport), HND is the primary airport servicing the Japanese domestic markets, and serves as the primary base of Japan's two major domestic airlines, Japan Airlines and All Nippon Airways. A disruption of this airport has, accordingly, significant implications for domestic and international connectivity for Tokyo. The second ranked airport is CAN (Guangzhou Baiyun International Airport), located in Guangzhou, the capital of China's Guangdong province (which is the country's highest GDP province). Interestingly, another airport in this region is among the top 10: SZX (Shenzhen Bao'an International Airport). Another Chinese airport takes rank 3: PEK (Beijing Capital International Airport), which was the second busiest airport worldwide before the COVID-19 pandemic, only behind ATL (Hartsfield–Jackson Atlanta International Airport). Moreover, all top-10 airports locate in either Asia or North America, but no airport in the third major aviation powerhouse, Europe. While North America, especially the United States, have a large number of airports, it seems like the airports in Europe are potentially better coordinated in terms of accessibility and schedule redundancies. In addition, there are various megapolis with high population densities in Asia, serving of which under disruptions can be a challenge; CGK (Soekarno–Hatta International Airport) is another prime example here.

The *GARITransfer* values from our planet-wide resilience analysis are reported in Fig. 15. The top-ranked airport is ATL (Hartsfield–Jackson Atlanta International Airport), which is the world's busiest airport for several years. A disruption at this airport has usually huge implications on the United States' domestic air transportation system and beyond. Notably, ATL is an important transfer hub in the United States and its hub carrier, Delta Airlines, is the world's largest air carrier. The fact that ATL ranked first for *GARITransfer* but does not appear in the top 10 of *GARIcommunity*, shows the ability of GARI to capture the resilience in terms of different aspects. The second ranked airport is DEN (Denver International Airport), a major hub for United Airlines and also one of the largest bases for Southwest Airlines and Frontier Airlines. Two more airports in the United States appear in the top 5: DFW (Dallas Fort Worth International Airport, which is the hub for American Airlines) and ORD (O'Hare International Airport, the hub for United and American). This indicates how a few hubs in the United States can be subject to major disruptions which then propagate through the air transportation system. The only non-US airport in the top 5 is IST (Istanbul International Airport), one of the two international airports serving Istanbul, Turkey. Despite only having opened in October 2018, IST is among the busiest airports in Europe. The airport succeeded ISL (Atatürk Airport) due to the latter's growth restrictions; and continues to play a central role for connections between Europe and Asia. Accordingly, a disruption at IST will have major impacts on inter-continental air connections.

Next, we investigate how closely *GARIcommunity* and *GARITransfer* values are correlated. Fig. 16 visualizes each airport as a circular marker with *GARIcommunity* represented along the x-axis and *GARITransfer* along the y-axis. Top-ranked airports are highlighted with their IATA codes. This figure further shows an absence of strong correlation between both indicators. Particularly, it can be seen that there are airports with a very high *GARIcommunity*, but significantly smaller *GARITransfer* values, e.g., LAS (Harry Reid International Airport) and MCO (Orlando International Airport), with several other airports located in China. Such airports either do not have a strong transfer function or are easily replaceable for such transfer functions during a disruption. The fact that the top-ranked airports concerning both resilience aspects are rather different has important policy implications. It highlights that — when protecting the airport infrastructures, one could focus on either accessibility/transfer aspects of the disrupted airports. This could, for instance, indicate that the airport should have a focused operation mode which puts an emphasis on either transfer or

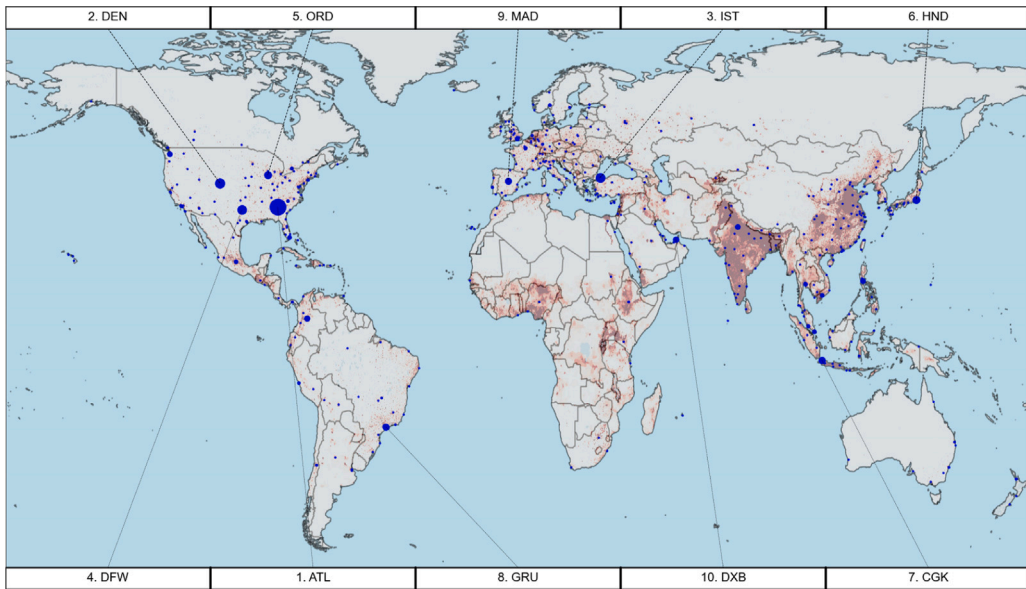


Fig. 15. GARItransfer results for the planet.

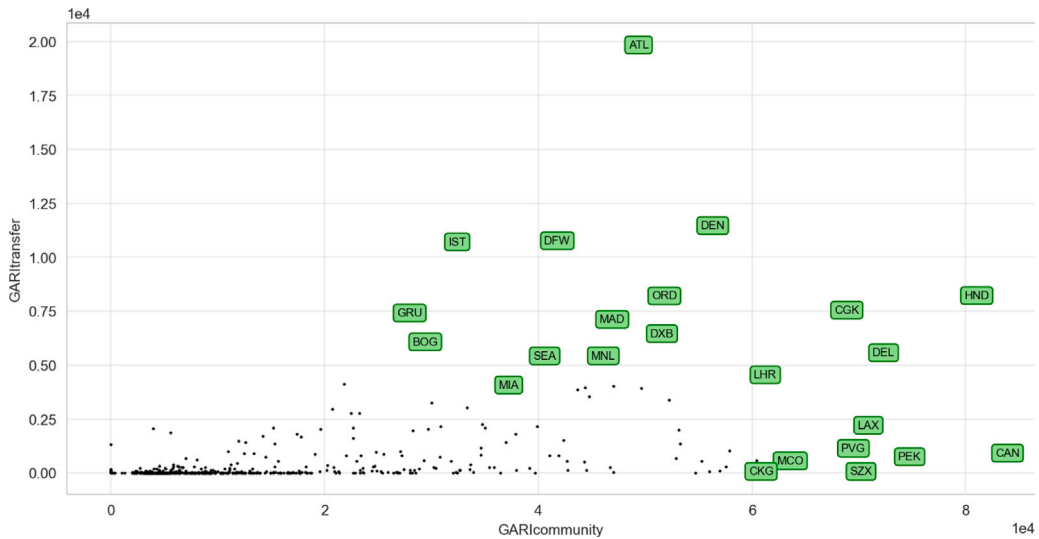


Fig. 16. Scatter plot of GARIcommunity (x-axis) and GARItransfer (y-axis) values.

community aspects. In this way, the limited resources during a disruption can be used more wisely and one can potentially achieve a quicker recovery.

In Fig. 17, we further explore on the correlation between GARI and some existing measures based on network science (DEG: Degree, BETW: Betweenness, CLOS: Closeness, PR: Pagerank). For each pair of methods, the Pearson correlation coefficient is reported by colour. The diagonal represents a Pearson correlation coefficient of 1.0. We find that neither of the four node importance metrics resembles either of the two GARI indicators closely.

Our final analysis in this study concerns the availability of transfer redundancies in the air transportation system. For this purpose, we collect statistics on the disrupted and recovered itineraries, identifying which airports frequently replace a specific disrupted airport. The results are shown in Fig. 18 (for North America), Fig. 19 (for Europe), and Fig. 20 (Asia), respectively. In each figure, the disrupted airport is shown along the y-axis and the alternate airport along the x-axis. The colour indicates how frequently (log-scale) the alternate airports are used instead of the disrupted airports. For each figure, we restrict the airports to the top-30 airports regarding total passengers served in 2023.

For instance, Fig. 18 reveals that a disruption at ATL leads to various opportunities for other airports to provide alternate travel options, mainly CLT (Charlotte Douglas International Airport), ORD (O'Hare International Airport), and DFW (Dallas Fort Worth

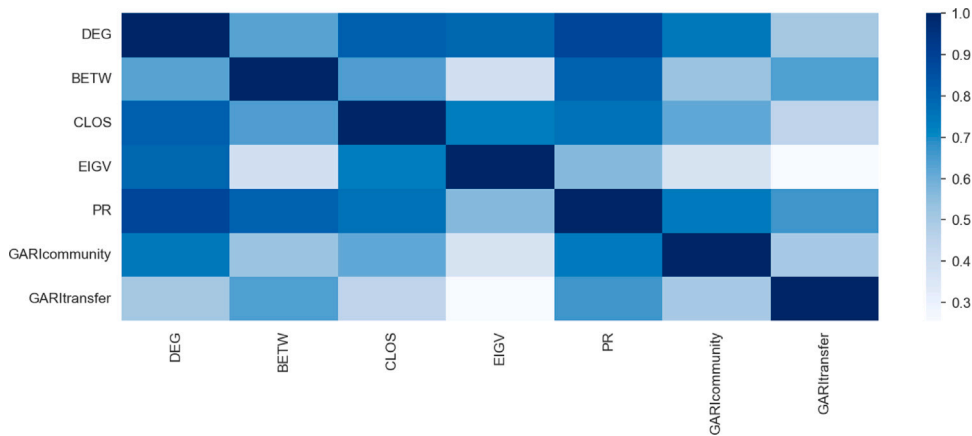


Fig. 17. Correlation analysis for *GARIcommunity* and *GARltransfer* against alternate methods.

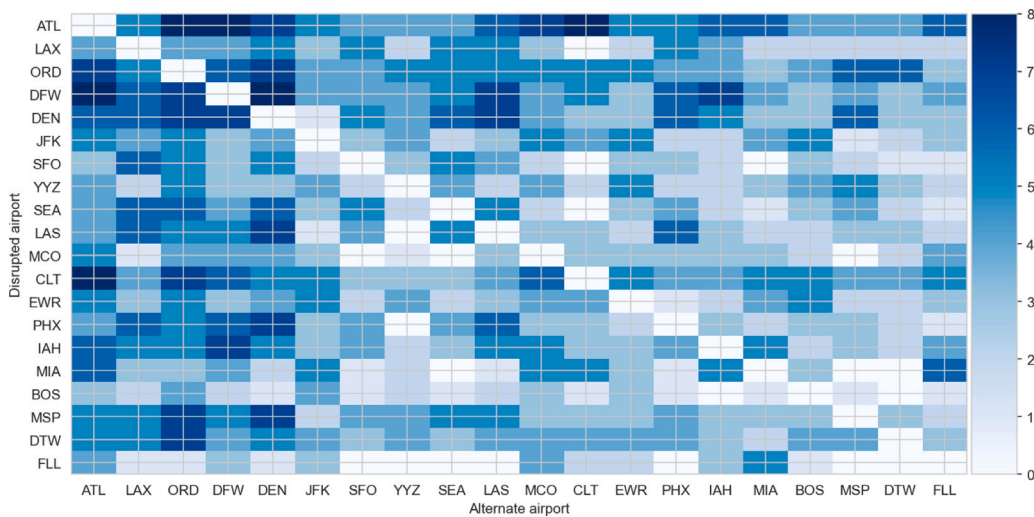


Fig. 18. Alternate airports for North America.

International Airport). But also many other airports provide alternatives; much more than in case of disruptions at other airports. Accordingly, one can conclude that a disruption at ATL is not so much about lack of capacity, but rather a matter of coordination how passengers are being rerouted and performing early ticket/resource allocation. Involved stakeholders should take this into account when conducting mitigation strategies.

Another interesting observation can be made from Fig. 19. IST (Istanbul Airport) and SAW (Istanbul Sabiha Gokcen International Airport) are highly complementary for transfer passenger services under disruptions. On one hand, this enables very high potential for contingency planning and effective rerouting, given the relatively close distance of 76 km (less than one hour driving). This means that passengers being routed through one of the airports could naturally be rerouted through the other airport without much deviation in travel time, given the similar flight trajectories involved. On the other hand, the spatial proximity of the two airports highlights the vulnerability of the system towards an outage of both airports at once, due to a spatial phenomenon, such as weather or political reasons. In this case, we could not find good recovery itineraries through other airports, leading to a potentially excessive disruption of the global transportation system. A similar observation can be made in Fig. 20 for the airport pair BKK (Suvarnabhumi Airport) and DMK (Don Mueang International Airport). Policy makers should pay closer attention to such potential cascades in the air transportation system. In this context it should be noted again, these results are for *GARltransfer*, not *GARIcommunity*. In the latter, the outage of two neighbored airports is clearly a disaster.

5. Conclusions

In this study, we have proposed a novel airport resilience measure, the Global Airport Resilience Index (GARI). While the existing measures in the literature usually have a strong focus on topological assessment, especially when executed at planet-scale, GARI is a simulation-based resilience indicator. The major contribution and findings of our study are summarized below:

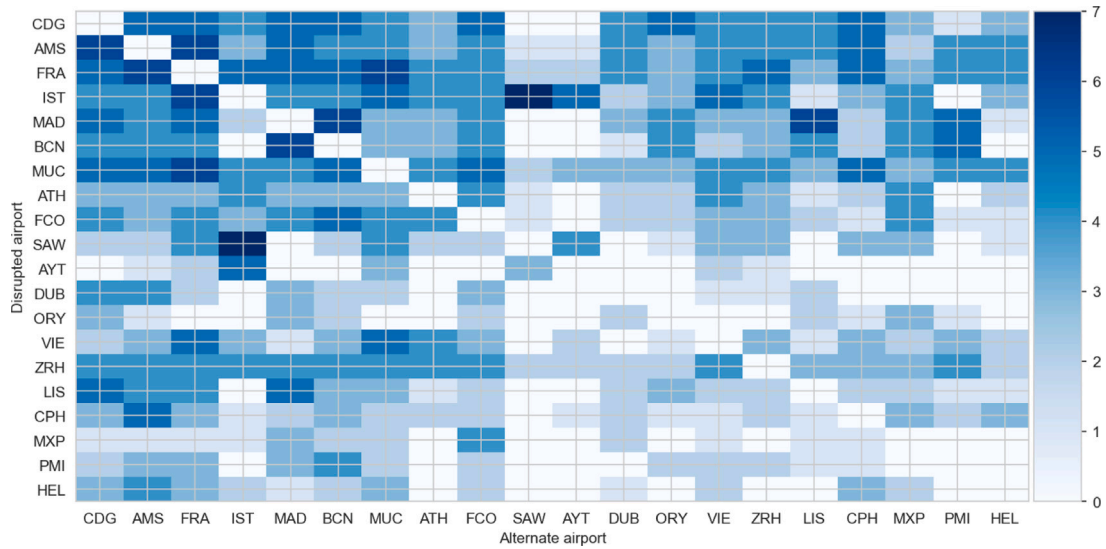


Fig. 19. Alternate airports for Europe.

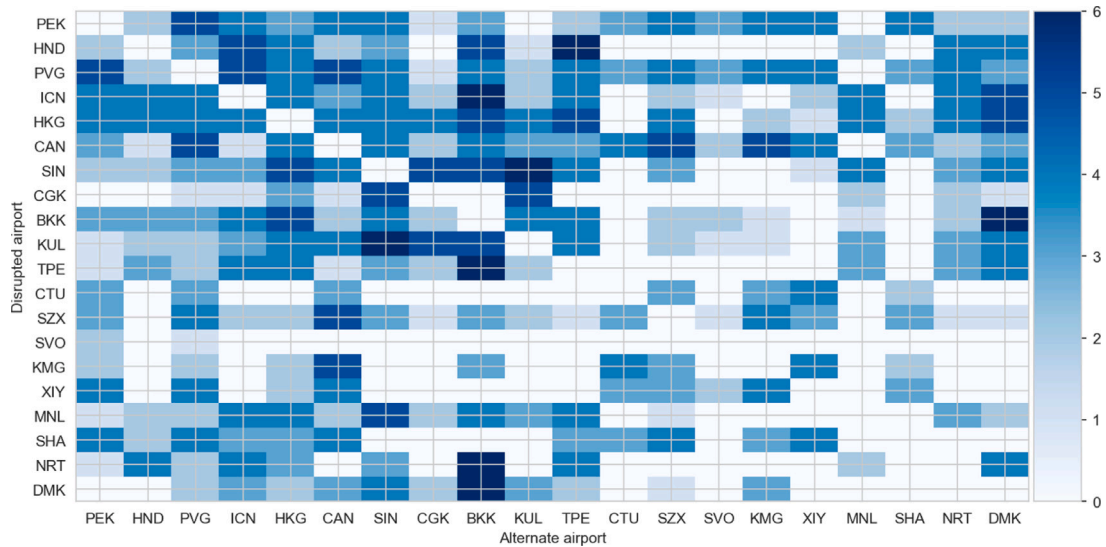


Fig. 20. Alternate airports for North Asia.

1. GARI is a realistic indicator which takes into consideration two distinct aspects of airport resilience: Community and transfer perspectives. Our experiments on worldwide air transportation data, population data and ground infrastructure transportation data reveal the feasibility/scalability of our index computation at the planet-scale, which is important to cover a complete picture of resilience, given the globally connecting nature of air transportation. Our experiments show that the two indicators indeed lead to rather different rankings, highlighting the need to assess different aspects of resilience and preparing adequate air transportation infrastructure protection and recovery strategies.
2. Concerning community resilience, our results indicate that the airport HND (Haneda Airport) serving Tokyo, CAN (Guangzhou Baiyun International Airport) serving Guangzhou/Greater Bay Area, and PEK (Beijing Capital International Airport) serving Beijing, are most important. We further find that all top-10 airports are located in either Asia or North America, but no airport in Europe, which is considered a major aviation market as well. While North America, especially the United States, have a large number of airports, it seems like the airports in Europe are potentially better coordinated in terms of accessibility and schedule redundancies. In addition, there are various megatropolis with high population densities in Asia, serving of which under disruptions can be a challenge. Following our experiments, CGK (Soekarno–Hatta International Airport) is a prime example for such a mega region.

3. Concerning transfer resilience, we find that ATL (Hartsfield–Jackson Atlanta International Airport) is the most important airport for transfer passengers. This finding is not really surprising, given that is presumably the most important transfer hub in the United States. With some gap, the next two most important airports are DEN (Denver International Airport) and IST (Istanbul International Airport). Two more airports in the United States appear in the top 5: DFW (Dallas Fort Worth International Airport) and ORD (O'Hare International Airport). This indicates how a few hubs in the United States can be subject to major disruptions which then propagate through the air transportation system.
4. Our experiments regarding correlations between *GARIcommunity* and *GARItansfer* as well as among other traditional resilience indicators reveals no strong correlations. This highlights the need to develop operation specific indicators for accurately measuring the resilience of the air transportation system towards disruptions at one airport. At the same time, this should also motivate other researchers to devise extended, realistic measures in future work. We believe that our work has laid the foundation for a scalable resilience framework; given that all experiments can be computed on commodity hardware within 1–2 weeks of pure simulation time; breaking through this bottleneck can be considered a major contribution of our study on its own.

Concerning the importance of GARI and relevance of our work to different air transportation stakeholders and communities, we summarize the major insights as follows:

1. GARI can inform the design and operational strategies of airports, becoming an essential tool for **airport managers and other airport-related stakeholders**. By understanding the potential impact of outages, airport operators are enabled to develop more robust infrastructure and operational procedures. For instance, if GARI shows a high reliance on a single airport, the airport could consider investing in additional capacity or improving the resilience of existing infrastructure to mitigate the risks of disruptions. Moreover, GARI can guide the development of emergency response plans. Airports with a higher GARI value, indicating greater potential impact from outages, might require more comprehensive contingency plans, including strategies for quickly rerouting passengers or providing alternative transportation options.
2. Another key application of GARI is its ability to identify vulnerabilities within the global air transportation system/network. By taking index values obtained through our GARI framework, **network-wide stakeholders** can pinpoint which airports are most susceptible to disruptions and which disruptions would have the most significant impact on both local and transfer passengers. For example, if GARI reveals that a particular airport is heavily relied upon by transfer passengers with limited alternative connection options, it highlights the need for contingency planning for that specific hub, particularly under circumstances where this hub is likely to be disrupted, e.g., due to upcoming weather events or staff outage. This knowledge allows regional/global air transportation authorities to prioritize investments (in the long run) as well as measures (in the short run) to bolster the resilience of these vulnerable airports.
3. Given that the air transportation system is a highly interconnected network where the resilience of individual airports affects the entire system, GARI can facilitate **better coordination among airports, airlines, and other stakeholders** by providing a common metric for resilience. With the existing measures for global air transportation network resilience, mostly based on relatively simple network science methodologies, most air transportation stakeholders had the consensus that they are unrealistic. Accordingly, our novel GARI can lie the foundation for a commonly used method. For instance, airports for which GARI indicates a strong mutual dependence might collaborate on shared contingency planning, such as mutual aid agreements where neighbouring airports support each other in the event of an outage. Airlines can use GARI to make more informed decisions about flight routes and schedules, ensuring that they minimize reliance on highly vulnerable airports.
4. GARI also has implications for **policy and regulation in the air transportation sector**. Regulators can use GARI to establish minimum resilience standards for airports or the entire air transportation system, based on a consistent methodology and global data, ensuring that all facilities meet a minimum threshold of resilience. This could lead to the development of resilience certification programmes, where regions or specific airports are rated based on their GARI score, incentivizing them to invest in resilience-enhancing measures. Additionally, GARI can inform public investment decisions. Governments can allocate resources more effectively by targeting regions and airports with low resilience scores, ensuring that public funds are used to enhance the overall resilience of the air transportation system.
5. Resilience is not a static quality but one that must undergo **continuous assessment and improvement**. GARI provides a framework for such an ongoing monitoring of airport resilience, given the availability of updated (a) OD passenger data and (b) grid-based population distribution data. By regularly updating GARI scores based on changes in the airport's infrastructure, operations, or the surrounding environment, stakeholders can ensure that they are always prepared for potential disruptions, despite changes to the underlying network structure or other modifications to the air transportation system. This continuous improvement process is critical in an era of increasing uncertainty, where airports face threats from natural disasters, terrorism, cyberattacks, and pandemics.

Our study comes with a set of limitations which are discussed next. These limitations, complementary with our major findings above, lead to a set of interesting and challenging research directions for future work. In this study, we have investigated the case of full airport closures. Another type of disruption is a partial closure, where the capacity of the airport is temporarily reduced, e.g., due to the outage of one runway or some events in a specific terminal (area). Developing a stochastic resilience measure which takes into consideration partial closures is an important future research direction, which presumably requires more sophisticated routing algorithms and some other simplifications might be required, unless the available hardware can compensate for the increase in computational requirements. In addition, our study is based on post-COVID-19 data. Comparing the resilience of

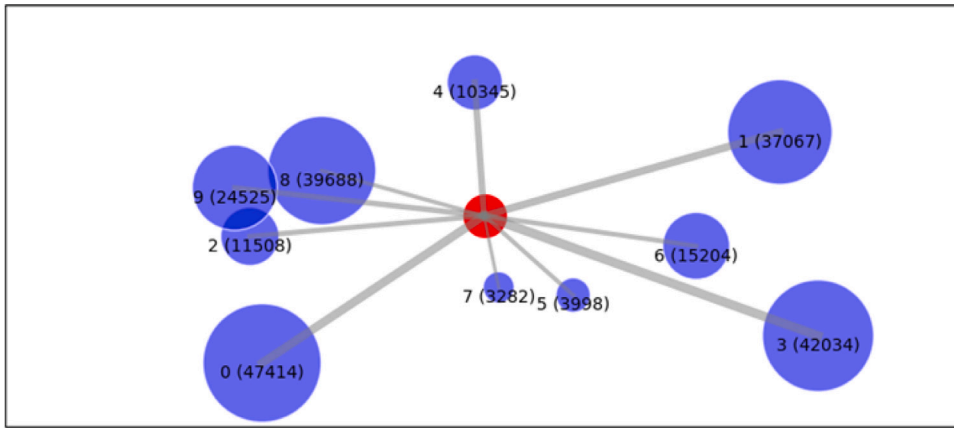


Fig. 21. Visualization of a toy example for normalization experiments.

airports before, during and post COVID-19 is another interesting aspect for future work. Finally, we have considered only limited ground infrastructure for travelling between airports. One mode that would be particularly important to investigate is (high-speed) rail. To date, to the best of our knowledge, there exists no global database of railway schedules which could be used for this purpose.

CRedit authorship contribution statement

Sebastian Wandelt: Writing – review & editing, Writing – original draft. **Anming Zhang:** Writing – review & editing, Writing – original draft. **Xiaoqian Sun:** Writing – review & editing, Writing – original draft.

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Appendix

Normalization

We have investigated nine distinct cases on a toy example. For this toy example, we have generated ten nodes around a centre node (representing the airport) with a random distance and random population, following a distribution obtained from the data in our experiments with global data. Fig. 21 visualizes this setup. All nodes are labelled with IDs from 0 to 9. The numbers of brackets represent the population at this node; for the sake of visualization, we did not investigate the more general case of grid cells here; the setup and results are, however, highly similar to this toy example. The width of lines in the figure represents the distance of the node from the centre node (airport).

Based on the toy example, we have computed nine quotients as follows, with the quotient label representing “Population/Distance” and four normalization functions (Orig: no normalization, Uni: all values set to 1, Min-Max: normalization to the range [0,1], and Log: computation of log values):

- Orig/Orig: Unnormalized population divided by unnormalized distance
- Orig/Uni: Unnormalized population divided by 1 (distance is eliminated)
- Uni/Orig: 1 divided by distance (population is eliminated)
- Orig/Min-Max: Unnormalized population divided by Min-Max normalized distance
- Min-Max/Orig: Min-Max normalized population divided by unnormalized distance
- Min-Max/Min-Max: Min-Max normalized population divided by Min-Max normalized distance
- Orig/Log: Unnormalized population divided by log(distance)
- Log/Orig: Log(population) divided by unnormalized distance
- Log/Log: Log(population) divided by log(distance)

The results of the relative node importance are visualized in Fig. 22 below. In each subplot, the size of a node corresponds to its importance with respect to a specific normalization scenario. The major insights can be summarized as follows:

- Orig/Orig gives a slight preference to Node 8 compared to Nodes 0/1/3, given that Node 8 is significantly closer than the other three highly populated nodes.

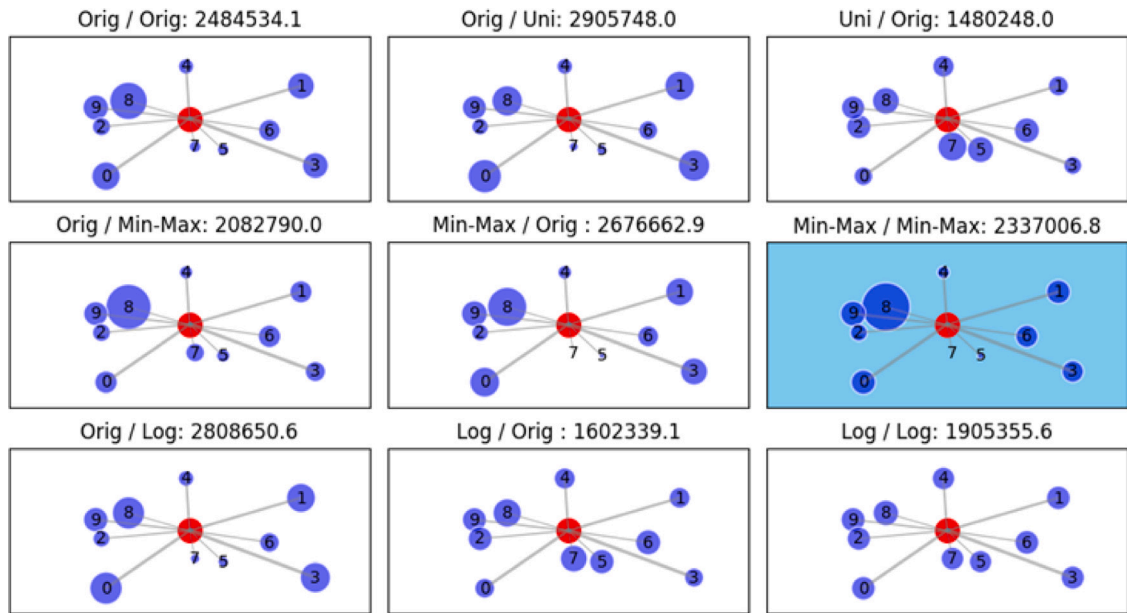


Fig. 22. Visualization of relative node importance.

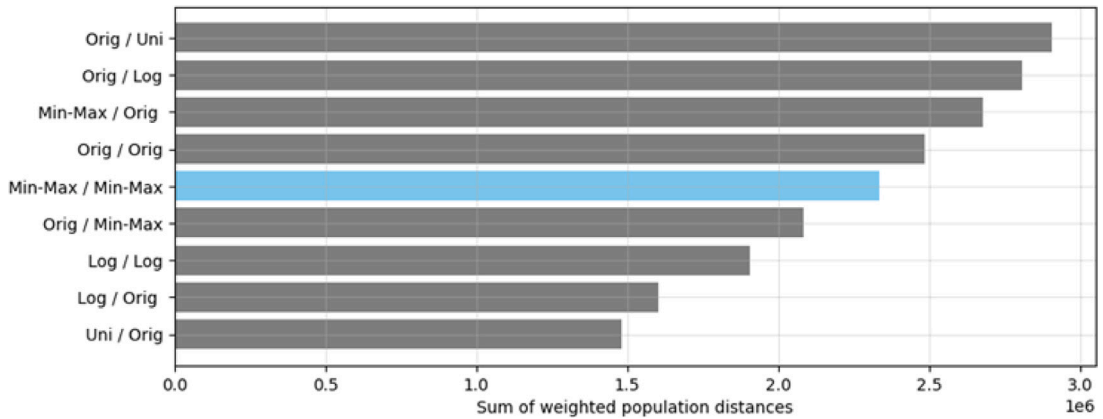


Fig. 23. Visualization of sum of weighted population distance.

- Orig/Uni assigns roughly equal importance to all four nodes 0/1/3/8. In addition, the importance of nodes with smaller populations is slightly lower than in Orig/Orig.
- Min-Max/Min-Max gives a very strong preference to Node 8, even stronger than Orig/Orig. In addition, nodes with small populations closer to the airport have very low importance. This aligns well with the intuition.
- All other cases either yield counter-intuitive results (Uni/Orig or Log/Log) or align well with one of the earlier three cases.

In order to further validate our choice of Min-Max/Min-Max, we have also plotted the obtained population distance scores for all nine measures as a bar plot in Fig. 23. Min-Max/Min-Max gives an “average” result away from border cases (e.g., Orig/Uni and Uni/Orig). Accordingly, we believe that Min-Max/Min-Max is a reasonable choice for our experiments. As a final verification means, we have also plotted the relative importance scores of all normalization measures as a pairwise scatter plot in Fig. 24. Min-Max/Min-Max (highlighted in blue) has a positive correlation with most other measures.

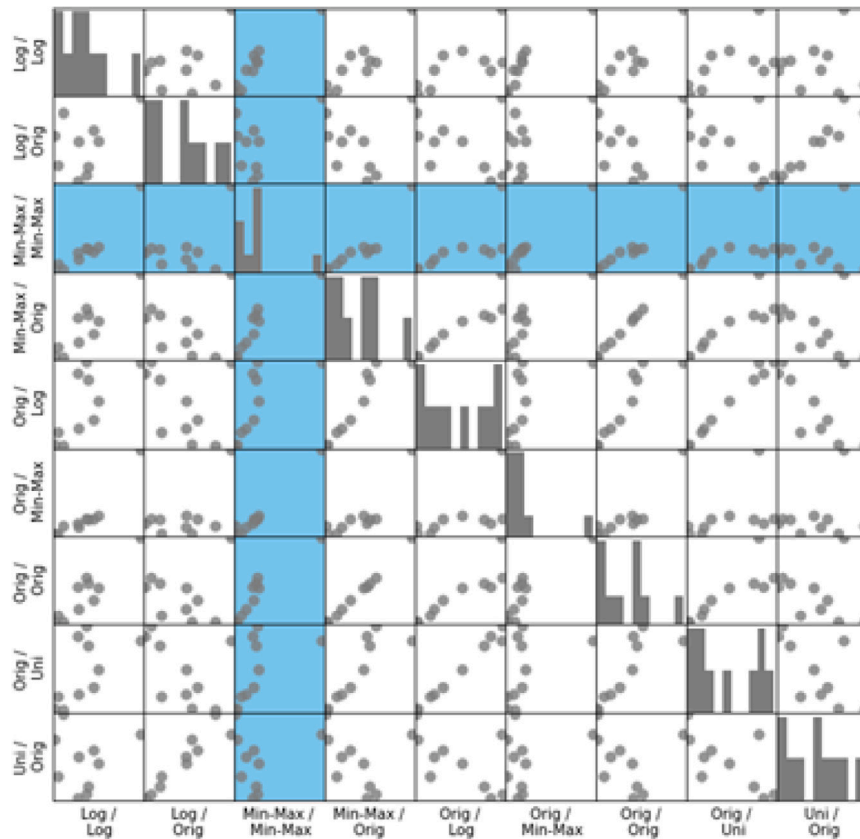


Fig. 24. Visualization of pairwise scatter plots concerning node importance.

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